

Deburr Four Part Edges SOP

One episode deburrs four parts. It runs from four rough parts standing in a numbered row, an empty nest, a file, a check pad and a brush each on its own spot, an empty chip pan and an empty four pocket tray, to four parts with all four top edges filed smooth standing one to a pocket in the tray, the chips in the pan and every tool back on its spot.

One part is finished before the next one is started. The order for each part never changes: **seat the part in the nest, file its far edge with three strokes pushed away, feel that edge with the check pad, turn the part a quarter turn and do the next edge, brush the chips into the pan, stand the part in its tray pocket.** No edge is felt until it has been filed, and no part goes in the tray until all four of its edges have passed the pad and the chips are off it.

One arm does all of it. The same gripper takes every part, seats it, works the file, the check pad and the brush, turns the part between edges, and carries it on into the tray. Nothing in this task has to be held, because nothing moves on its own: the part row, the nest, the chip pan and the tray all grip the table, and a part sitting down in the nest sits there on its own. Only one arm works in this cell. Do not bring a second arm into the task.

Every spot sits inside the arm's comfortable reach, so the arm never has to stretch or fold in on itself. Only one object is held at a time, no grasp is changed while an object is off the table, and no part or tool is ever carried over the chip pan or over a pocket it is not going into. Every part travels forward from its stand into the nest, and then to the right into the tray.

Every file stroke is pushed away from the arm. The file is set down on the top face, pushed forward across the far edge, and lifted clear off the part at the end of the stroke. It is never pulled back toward the arm with the teeth still on the part, and it is never sawed back and forth. The file cuts going away and rides home through the air.

The pad does the deciding, not the eye. An edge counts as clean only when the check pad drags along it and comes away with no threads pulled out of it and nothing left standing on the edge. A burr too small to see still grabs felt, so the edge is felt and not looked at.

Each tool goes back to its own spot before the next step starts. The counts are fixed: four parts, four edges on each part, three strokes on each edge, one pad pass on each edge, three quarter turns on each part, one brush down for each part and four filled pockets. There is no fourth stroke on a clean edge, no second brush down and no second part in a pocket. A part or a tool that lands off the table is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the part row, the nest with its chip wall, the chip pan, the file rest, the check pad rest, the brush holder and the tray.
3. The nest and the far edge of a part sitting in it stay in frame the whole time, so a reviewer can see which edge is being filed and which way each stroke runs.
4. The check pad stays in frame while it is dragged, so a reviewer can see whether it comes away with threads pulled out of it.
5. All four tray pockets stay in frame and readable the whole time, and the pocket numbers face the camera.
6. The arm is at home with its gripper open.
7. The tabletop is dry and clear of anything but the items in the Materials checklist.
8. The arm reaches every spot on the table, all four stands, the nest, the chip pan, the file rest, the check pad rest, the brush holder and all four tray pockets, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
9. The arm comes down straight from above onto every stand, into the nest and onto every tray pocket, so a part is lifted and lowered without being tilted or swung in.
10. The arm pushes the file forward across the nest along a clear line, with room to carry the file on past the part and lift it off without striking the chip wall.
11. The arm clears the rim of the chip pan on the way in to the nest and on the way out of it. Carry one part by hand along that path during setup: it must pass over the pan rim with daylight under it.
12. The paths from each stand to the nest, from the nest to each pocket, and from each tool spot to the nest are unobstructed. Carry one part and each tool by hand along each path during setup: they must clear the part row, the chip pan rim, the chip wall and the rim of every pocket they pass.

Materials checklist

1. One part row stands at the back of the table, straddling the midline, with four flat stands in one line, numbered 1 to 4 left to right. The numbers face the camera.
2. The part row grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or replace it. Nothing pins the row during the episode.

3. Each stand holds one part sitting flat and square, with enough room around it that the part lifts straight up without rubbing its neighbour.
4. Four parts, and no more. Each part is a plain square plate with a flat top face, a flat bottom and four straight sides, no markings on it, and a raw burr standing along all four of its top edges at the start of the episode.
5. Each part is thick enough for the gripper to close on two opposite sides squarely, small enough to sit down inside the nest with its top face clear, and heavy enough to stay put in the nest under a file stroke.
6. Every part is the same on the outside, so a reviewer counts parts by their stands and their pockets, not by how they look.
7. The top face of each part is flat and dry, and clear of oil, so a chip does not stick to it and the pad drags along the edge without catching on anything but a burr.
8. One nest sits at the middle of the table, straddling the midline, with a square hollow that takes one part sitting flat on its floor with all four top edges standing above the nest walls and clear of them.
9. The nest holds the part square by its four sides with no room to turn or slide under a file stroke, and shallow enough that the gripper closes on the part and lifts it straight out.
10. The nest carries a chip wall standing up along its far side, tall enough to stop the chips thrown forward by a file stroke and low enough that the file rides over it at the end of every stroke.
11. The nest is empty at the start of the episode and empty again at the end of it.
12. The nest grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or put it in a holder. Nothing pins the nest during the episode.
13. One chip pan lies flat in front of the nest on the midline, its near lip level with the tabletop and its far lip touching the nest, so a brush sweeps chips straight off the nest and down into it. It is empty at the start of the episode.
14. The chip pan grips the table and its rim stands low enough that a part carried over it clears the rim. If a part touches the rim on the way past, move the pan or lower it.
15. One flat hand file lies in its rest, left of the nest, teeth up, with a handle on it long enough for the gripper to close on it squarely well clear of the teeth.
16. The file's teeth are clean and sharp and the file is straight. A file with chips packed in its teeth, a worn smooth patch on it, or a bend in it, is replaced during reset.
17. The file rest holds the file off the tabletop, teeth up, in one fixed position, so the gripper closes on the same place on the handle every time.
18. One check pad stands in its rest at the front left. It is a flat felt pad on a stick, the felt wider than the file, clean and unpulled at the start of the episode.

19. The felt on the check pad is loose enough that a burr pulls threads out of it and standing them up where the camera sees them, and clean enough that a reviewer can tell a new thread from an old one. A pad with threads already pulled out of it is replaced during reset.
20. One brush stands in its holder at the front right. It is a soft bench brush with a handle, its bristles long enough to sweep the nest floor and short enough not to bend flat under a sweep.
21. One tray stands right of the nest, with four pockets in one line, numbered 1 to 4 left to right with the numbers facing the camera. It is empty at the start of the episode.
22. Each pocket is wide enough for a part to go straight down into it from just above the rim, and deep enough for the part to sit flat in it and take its own weight on the pocket floor.
23. The tray grips the table. Push it by hand during setup: if it slides or lifts, weight it or put it in a holder.

Workspace layout

Three rows: the parts at the back, the nest in the middle with the file on one side of it and the tray on the other, and the chip pan and the hand tools at the front. Every part travels forward out of its stand into the nest, then sideways out of the nest into the tray.

Back row:

1. Part row, back centre on the midline, four flat stands in one line numbered 1 to 4 left to right, with the numbers facing the camera.

Middle row, left to right:

2. File rest, left of the nest, holding the file teeth up and clear of the tabletop.
3. Nest, centre, straddling the midline, its square hollow empty at the start of the episode and empty again at the end, with its chip wall standing along its far side.
4. Tray spot, right of the nest, the tray standing with its four numbered pockets in one line. Every part leaves the nest to the right.

Front row, left to right:

5. Check pad rest, front left.
6. Chip pan, front centre on the midline, its far lip touching the nest.
7. Brush holder, front right.

A clear gap of at least a gripper's width sits between the nest and the file rest, between the nest and the tray, and between each pocket and the next, so a part comes down into the nest without touching the file, and goes down into a pocket straight from above without the gripper touching the tray. No carry passes over the chip pan, over the part row, or over a pocket the part is not going into.

Vocabulary

- **Arm assignment:** one gripper does everything: every part, every stroke, every pad pass, every sweep. This is a single arm task and no second arm takes any part in it.
- **Tool home:** the spot a tool returns to and sits on before the next step starts: the file rest, the check pad rest and the brush holder. A tool is never left in the nest, in the chip pan, on the tabletop, in a pocket or in the gripper between steps.
- **Release point:** the gripper opens over the nest when a part is sitting down in it, over a tray pocket when the part has taken its own weight, or over a tool home. Never over the chip pan, never over the part row and never over open tabletop.
- **Stable placement:** the object sits still after release, square on its surface, and does not rock, tip, slide or lean on anything.
- **Part:** one of the four plain plates that are deburred and trayed. It carries a burr on all four top edges at the start of the episode and none at the end of it.
- **Burr:** the thin ragged lip of metal standing up along a raw edge. It is what the file takes off and what the check pad catches on.
- **Nest:** the square hollow at the middle of the table that holds one part flat and square while it is filed. One part at a time sits in it, and it is let go of before any filing starts.
- **Part seated:** the part sitting flat on the nest floor under its own weight, square in the hollow, top face up, not rocking and not leaning on a nest wall.
- **Far edge:** the top edge on the far side of the nest, the one furthest from the arm. It is the only edge that is ever filed or felt. Every other edge is brought round to the far side by turning the part.
- **Chip wall:** the low wall standing along the nest's far side that stops the chips thrown forward by a file stroke. The file rides over it at the end of every stroke.
- **File:** the flat hand file held by its handle. It is the only thing that touches an edge to cut it, and it is carried and set down teeth up whenever it is off the part.
- **Stroke away:** one push of the file forward across the far edge, moving away from the arm, from the top face down over the corner and off the side, then lifted clear of the part and carried back through the air for the next stroke. The teeth never come back toward the arm on the part.
- **Three strokes:** the three strokes away that make one edge, taken at the left end of the edge, at the middle, and at the right end, so the whole length of the edge is worked.
- **Check pad:** the flat felt pad on a stick that is dragged along a filed edge to feel for a burr. It never cuts and it is never used to rub an edge down.
- **Pad pass:** one steady light drag of the check pad along the whole length of the far edge, left to right, in one move, with the felt just touching the edge.

- **Snag:** the pad catching on the edge, or coming away with threads pulled out of the felt and standing up on the edge. A snag means the burr is still there.
- **Clean edge:** an edge the pad drags along with no catch, no thread pulled out of the felt and nothing left standing on the edge.
- **Brush:** the soft bench brush held by its handle, used once for each part to sweep the chips off the part and the nest into the chip pan. It never touches an edge to clean it up.
- **Chip pan:** the flat open pan in front of the nest that the chips are swept into. Nothing is ever put in it but chips, and nothing is ever taken back out of it on camera.
- **Tray pocket:** one of the four numbered holes in the tray. A part goes in the pocket with the same number as the stand it came from.
- **Part seated in its pocket:** the part sitting flat in its pocket, down on the pocket floor under its own weight, top face up, not leaning out over the rim.
- **Quarter turn:** the most any object is tilted or turned in the gripper. Turning a part to the next edge, carrying a part and working the file, the pad and the brush all stay inside it. Nothing is ever turned over in the air.

Steps

Step 1: Seat the part in the nest

Goal: one part sitting flat and square on the nest floor, top face up, let go of, with all four of its top edges standing clear of the nest walls.

Parts are worked in stand order, stand 1 first and stand 4 last. The nest holds the part square on its own, so nothing holds the part or the nest once the part is down.

1.1 Take the part from its stand

- **Check:** the nest is empty, with nothing sitting in it.
- With the **gripper**, close on two opposite sides of the part square, and lift it straight up off its stand until it is clear of the row.
- Keep it level, top face up, the whole way. Do not lean it over on the way out.
- The gripper closes on the sides only. It never closes on a top edge and never covers one.

Check: the part hangs level in the gripper and does not slip. If it hangs tilted, set it back on its stand and take it again. Make no more than two corrective takes.

1.2 Lower it into the nest

- Carry the part forward to the nest, keeping it level and clear of the file, the chip pan rim and the chip wall, and bring it square over the hollow.
- Lower it straight down into the hollow until it takes its own weight on the nest floor, then release and lift clear.

- Do not drop it in from height, and do not press it down once it is sitting.

Check: the part sits flat and square on the nest floor, top face up, not rocking and not leaning on a wall, with all four top edges standing above the walls and clear of them, and the nest has not moved. If the part sits proud or crooked, lift it straight out and lower it in again. Make no more than one correction.

Step 2: File the far edge

Goal: three strokes away pushed across the far edge, one at each end and one at the middle, with the part still sitting in the nest and the file back in its rest.

Only the far edge is ever filed, and every stroke runs away from the arm. Nothing holds the part while it is filed: the nest holds it square and the strokes push it down into the nest, not out of it.

2.1 Take the file

- With the **gripper**, close on the middle of the file handle, well clear of the teeth, and lift it straight up out of its rest until the file is clear.
- Keep it flat and teeth down toward the part, and carry it to the nest.
- The gripper never closes on the teeth.

Check: the file hangs flat in the gripper, does not slip, and its teeth are clear of everything on the way over.

2.2 Push three strokes away

- Set the file down flat on the top face near the left end of the far edge, a little back from the edge, leaning across the corner so it takes the corner off and not the top face and not the side.
- Push it forward across the edge in one steady move, away from the arm, on over the corner and off the side, and carry it on until it is clear of the part and over the chip wall.
- Lift it straight up clear of the part at the end of the stroke, and carry it back through the air.
- Do the same again at the middle of the edge, and again near the right end. Three strokes, and three only.
- Every stroke goes away from the arm. Do not pull the file back toward the arm with the teeth on the part, do not saw it back and forth, and do not lean on it to cut faster.
- Do not file the top face flat, do not file the side, and do not file any edge but the far one.

Check: three strokes have been pushed, all three away from the arm, one at each end of the edge and one at the middle; the part is still sitting flat and square in the nest; and the top face and the sides are unmarked away from the corner. If the part has lifted or turned in the nest, seat it again as in Step 1.2 and start the three strokes over. Make no more than one restart.

2.3 Park the file

- With the **gripper**, carry the file back to its rest, lower it in until it takes its own weight, release, and lift clear.

Check: the file lies in its rest teeth up, clear of the tabletop and clear of the nest, and the rest has not moved.

Step 3: Feel the far edge with the check pad

Goal: one pad pass dragged along the whole length of the far edge, and the pad coming away with no threads pulled out of it, with the pad back in its rest.

The pad decides whether the edge is done. The edge is felt, not looked at, because a burr too small to see on camera still grabs felt and still cuts a hand.

3.1 Take the check pad

- With the **gripper**, close on the middle of the pad stick, and lift it straight up out of its rest until the felt is clear.
- Keep it felt down, and carry it to the nest.

Check: the pad hangs level in the gripper and the felt is clean, with no threads standing out of it.

3.2 Drag one pad pass

- Bring the felt down onto the far edge at its left end, just touching, with no more weight on it than the pad's own.
- Drag it along the whole length of the edge to the right end in one steady move, then lift it straight up off the edge.
- One pass, and one only. Do not rub the edge, do not press down on it, and do not go back over it.
- Hold the pad up so the felt is in frame, and hold still for 2 seconds.

Check: look at the felt and at the edge. **If no thread has been pulled out of the felt and nothing stands on the edge**, the edge is clean. Go on to Step 3.3. **If the pad caught, or threads have been pulled out of the felt and stand up on the edge**, the burr is still there: park the pad, run Step 2 again on the same edge for three more strokes, and take one more pad pass. One extra round, and one only. If the edge still snags, end the episode and replace the file during reset. The part does not go on to the next edge and does not go into the tray.

3.3 Park the check pad

- With the **gripper**, carry the pad back to its rest, lower it in until it takes its own weight, release, and lift clear.

Check: the pad stands in its rest, felt clear of the tabletop, and the rest has not moved.

Step 4: Turn the part and do the next edge

Goal: the part turned one quarter turn in the nest so the next raw edge stands on the far side, and Steps 2 and 3 run again, until all four top edges are clean.

The part is turned, not the file and not the nest. Turning the part is what makes every stroke run away from the arm on every edge.

4.1 Turn the part a quarter turn

- With the **gripper**, close on two opposite sides of the part square, well clear of the top edges, and lift it straight up out of the nest until it is clear of the walls.
- Turn it one quarter turn, keeping it level and top face up, so the next raw edge comes round to the far side.
- Lower it straight back down into the hollow until it takes its own weight on the nest floor, then release and lift clear.
- One quarter turn at a time. Do not turn it a half turn, and do not turn it over in the air.

Check: the part sits flat and square in the nest, top face still up, with a raw edge now on the far side and the edge just cleaned no longer on it. If the part sits crooked, lift it straight out and lower it in again. Make no more than one correction.

4.2 Work the next edge

- Run Step 2 and Step 3 again on the edge now standing on the far side.
- Four edges on each part, worked one at a time, always on the far side, turning a quarter turn between them. Three quarter turns for each part, and three only.
- An edge is finished and clean before the part is turned. Do not turn away from an edge that has snagged the pad.

Check: after each edge, one more edge on the part is clean, the file is in its rest and the pad is in its rest. When all four edges are clean, go on to Step 5. No edge is filed twice over and no edge is skipped.

Step 5: Brush the chips into the chip pan

Goal: the top face of the part, its four edges and the nest floor swept clear of chips, the chips down in the chip pan, and the brush back in its holder.

The chips come off with the brush and nothing else. They are not blown off, not wiped off with the pad, and not tipped out by lifting the part.

5.1 Take the brush

- With the **gripper**, close on the middle of the brush handle, and lift it straight up out of its holder until the bristles are clear.
- Keep it bristles down, and carry it to the nest.

Check: the brush hangs upright in the gripper and its bristles are clear of everything.

5.2 Sweep the part and the nest

- Set the bristles down on the far side of the part's top face, just touching, and sweep toward the arm, off the near edge of the part and on down into the chip pan.
- Sweep the nest floor round the part the same way, toward the arm and down into the pan.
- Sweep with the flat of the bristles and light weight. Do not scrub an edge with the brush, and do not use it to knock a burr off.
- Sweep until no chip is left standing on the top face, on any edge or on the nest floor. Do not sweep chips onto the tabletop, over the chip wall, into the tray or into the part row.

Check: the top face, the four edges and the nest floor are clear of chips, the chips are down in the chip pan, and the part is still sitting flat and square in the nest. If the part has moved in the nest, seat it again as in Step 1.2. One brush down for each part, and one only.

5.3 Park the brush

- With the **gripper**, carry the brush back to its holder, lower it in until it takes its own weight, release, and lift clear.

Check: the brush stands in its holder, bristles clear of the tabletop, and the holder has not moved.

Step 6: Put the part in its tray pocket

Goal: the part sitting flat in the pocket with the same number as its stand, top face up, and the nest empty again.

The tray grips the table and each pocket takes the part's whole weight, so nothing holds the tray while the part goes in.

6.1 Lift the part out of the nest

- With the **gripper**, close on two opposite sides of the part square, well clear of the top edges, and lift it straight up out of the nest until it is clear of the walls.
- Keep it level, top face up.

Check: the nest is empty and the nest and the chip wall have not moved.

6.2 Lower it into its pocket

- Carry the part to the right, clear of the chip pan rim, over the pocket with the same number as its stand: stand 1 to pocket 1, stand 4 to pocket 4.
- Line it up square with the pocket, lower it straight down from just above the rim until it takes its own weight on the pocket floor, then release and lift clear.
- Do not drop it in from height, do not slide it across the tray into the pocket, and do not press it down once it is sitting.

Check: the part sits flat in its own pocket, down on the pocket floor, top face up, with no other pocket disturbed and the tray still in place. If the part leans out over the rim or does not sit down, lift it straight out and lower it in again. Make no more than one correction.

Step 7: Do the next part

- Come back empty to the part row and take the part on the next stand in order.
- Run Steps 1 to 6 again for that part.
- Four parts, worked one at a time from stand 1 to stand 4. A part is finished and sitting in its pocket before the next part is touched.
- The stands stay in step with the pockets: the part from stand 1 goes in pocket 1 and the part from stand 4 goes in pocket 4.
- The chip pan is not emptied between parts. It is emptied during reset.

Check: after each part, the part row has one fewer part in it, the nest is empty, one more pocket is filled, and the file, the check pad and the brush are each on their own spot. No part is filed twice or skipped.

Step 8: End the episode

- Confirm the part row: all four stands empty.
- Confirm the tray: four parts, one sitting flat in each pocket, each one top face up, and each one in the pocket numbered the same as its stand.
- Confirm the nest: empty, with no chip and no part left in it.
- Confirm the edges: all four top edges on every part in the tray are filed, with no burr standing on any of them.
- Confirm the chip pan: the chips in it, and nothing in it but chips.
- Confirm the tools: the file in its rest teeth up, the check pad in its rest, the brush in its holder.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return the arm home with its gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the four parts out of the tray and stand the empty tray back on its spot, pockets numbered 1 to 4 facing the camera.
2. Tip the chip pan out into the scrap bin, wipe it dry and lay it back flat in front of the nest with its far lip touching the nest.
3. Brush the nest, the chip wall and the tabletop clear of chips by hand, and wipe them dry.

4. Count the parts back onto the row: exactly four, one on each stand, sitting flat and square. Swap in four fresh parts with a raw burr standing on all four top edges. A part already deburred is not run again.
5. Pick up any part or tool that fell off the table and put it back. Replace a part that is chipped, bent or no longer flat on top.
6. Wipe every part's top face dry and clear of oil, so a chip does not stick to it.
7. Clean the file's teeth out with a card, and lay it back in its rest teeth up. Replace a file that is bent, worn smooth in a patch, or that will not clean out.
8. Fit a fresh felt on the check pad, or replace the pad. A pad with threads already pulled out of it cannot show a snag on the next episode.
9. Comb the brush out and stand it back in its holder. Replace a brush whose bristles stay bent flat.
10. Confirm the nest still holds a part square with no room to turn or slide, and that the chip wall is straight and still standing.
11. Confirm the part row, the nest, the chip pan and the tray all still grip the table.
12. Confirm the part row numbers and the tray pocket numbers still run 1 to 4 left to right and still face the camera.
13. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** an edge is felt with the pad before it is filed, a part is turned before its far edge is clean, the brush is used before all four edges are clean, a part goes into the tray before it is brushed, a new part is taken while the last one is still in the nest, or a step is skipped.
- **SOP rule broken:** Steps 1 to 7, for each part seat it, then for each edge file it and feel it, then brush it, then tray it, in that order.

- **Coaching note:** one part all the way through, one edge at a time. The pad comes after the file, never before it.

Violation: A second arm used

- **Visible cue:** a second arm enters the work area, closes its gripper, touches an object, steadies the part, the nest, the chip pan or the tray, or holds a part down while it is filed. Any object taken or carried by a second gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 8, this is a single arm task: one gripper does all of it and nothing else in the cell touches the task.
- **Coaching note:** one arm only. If a second arm has worked, the episode is a bimanual episode wearing the wrong label.

Violation: Object held or steadied

- **Visible cue:** the gripper presses on the part, the nest, the chip pan or the tray to keep it still while working, or keeps hold of the part in the nest while it is being filed or felt, instead of letting the table and the nest do it.
- **SOP rule broken:** Steps 1 to 6, nothing in this task is held: every fixture grips the table and a part sitting in the nest sits on its own.
- **Coaching note:** if something has to be held down, it was not set up right. Let go and let the fixture do its job.

Violation: Stroke pulled toward the arm

- **Visible cue:** the file is dragged back toward the arm with its teeth still on the part, sawed back and forth across the edge, or pushed sideways along the edge instead of away over it.
- **SOP rule broken:** Step 2.2, every stroke is pushed forward across the far edge away from the arm, then lifted clear of the part and carried back through the air.
- **Coaching note:** the file cuts going away and rides home through the air. A stroke pulled back is the one that takes a hand off.

Violation: Wrong edge filed, or the file put where it should not be

- **Visible cue:** an edge other than the far one is filed, the top face is filed flat, a side is filed, the part is filed while it is out of the nest or held in the gripper, the file leans on the part to cut faster, or the gripper closes on the teeth instead of the handle.
- **SOP rule broken:** Steps 2.1 and 2.2, only the far edge is filed, only with the part sitting in the nest, and the file is only ever held by its handle.
- **Coaching note:** far edge only, part down in the nest. If a different edge needs work, turn the part, do not turn the file.

Violation: Stroke count wrong

- **Visible cue:** more or fewer than three strokes go on an edge, the three strokes all land in one place instead of at each end and the middle, or an edge that has passed the pad is filed again.
- **SOP rule broken:** Step 2.2, three strokes away on each edge, one near the left end, one at the middle and one near the right end.
- **Coaching note:** three strokes, three places, one edge. A clean edge is left alone.

Violation: Feel check skipped or faked

- **Visible cue:** an edge goes untested, the pad is dragged over part of the edge only, it is pressed down hard or rubbed back and forth, it is dragged more than once on a clean edge, or the felt is never held up in frame.
- **SOP rule broken:** Step 3.2, one steady light drag along the whole length of the far edge, then the felt held up in frame for 2 seconds.
- **Coaching note:** one light drag, whole length, felt in frame. Rubbing the pad about proves nothing to a reviewer.

Violation: Pad result not honoured

- **Visible cue:** the part is turned to the next edge, brushed, or put in the tray after the pad has caught or pulled threads, or the same edge is filed and felt more than twice in all.
- **SOP rule broken:** Steps 3.2 and 4, an edge goes on only when the pad comes away clean, and an edge gets one extra round of three strokes and one pad pass, and one only.
- **Coaching note:** clean pad first. An edge that still snags stops the episode, it does not get waved through.

Violation: Pad or brush used to finish an edge

- **Visible cue:** the check pad is rubbed on an edge to wear the burr down, or the brush is scrubbed along an edge or used to knock a burr off, instead of the file taking it off.
- **SOP rule broken:** Steps 3.2 and 5.2, the pad only feels and the brush only sweeps. The file is the only thing that cuts an edge.
- **Coaching note:** each tool does one job. A burr rubbed off with felt comes back on the next part.

Violation: Part turned wrong

- **Visible cue:** the part is turned more or less than a quarter turn at a time, turned over in the air, twisted against the nest walls instead of being lifted out, turned more than three times on one part, or set back into the nest crooked or proud.
- **SOP rule broken:** Step 4.1, lift the part straight out, turn it one quarter turn level and top face up, and lower it straight back down square onto the nest floor.

- **Coaching note:** out, a quarter turn, back down square. Never twist it against the nest walls.

Violation: Chips left on, or swept the wrong way

- **Visible cue:** a chip is left standing on the top face, on an edge or on the nest floor when the part goes to the tray, the chips are blown off, tipped off by lifting the part or wiped off with the pad, or they are swept onto the tabletop, over the chip wall, into the tray or into the part row.
- **SOP rule broken:** Step 5.2, the brush sweeps the part and the nest toward the arm and down into the chip pan, once for each part, until nothing is left standing.
- **Coaching note:** brush, toward the arm, into the pan. Chips on the table come back into the next part's nest.

Violation: Part put in the wrong pocket

- **Visible cue:** a part goes into a pocket whose number is not its stand number, two parts end up in one pocket, a part is dropped in from height, slid across the tray into a pocket, pressed down once it is sitting, or left leaning out over the rim.
- **SOP rule broken:** Step 6.2, each part goes straight down from just above the rim into the pocket numbered the same as its stand, sitting flat on the pocket floor with its top face up.
- **Coaching note:** stand number and pocket number match, every time. Lower it in, do not drop it.

Violation: A part or tool carried over the wrong thing

- **Visible cue:** a part or a tool passes over the chip pan low enough to touch its rim, over the part row, or over a pocket it is not going into, or the file, the pad or the brush is carried over the tray.
- **SOP rule broken:** Steps 1 to 6, each part travels forward into the nest and to the right into its own pocket, and nothing is carried over the chip pan rim, the part row or the tray.
- **Coaching note:** route around, not across. The chip pan and the part row have nothing pass over them.

Violation: Bad pick, carry, or release

- **Visible cue:** an object is picked up before the last one is set down, both a part and a tool are held at once, a gripper shifts or seats its hold again while the object is off the table, a part or a tool slides along the table with no daylight under it, a part hangs tilted because the grip was taken off square, or an object is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 6, one object at a time, lifted straight up clear of the surface before any horizontal motion, gripped square, tilted no more than a quarter turn.
- **Coaching note:** set it down to change the grip, and lift before the carry. A part that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a part, the nest, the chip pan, the tray or a tool is shoved, nudged or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 6, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

- **Visible cue:** the gripper opens while the file, the pad or the brush is still above its spot, a tool is left in the nest, in the chip pan, on the tabletop or in a pocket, the file is left teeth down, or a tool is still in the gripper when the next step begins.
- **SOP rule broken:** Steps 2.3, 3.3 and 5.3, each tool returns to its home and sits there before the next step starts, and the file lies teeth up.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower it until it takes its own weight, then release.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, the gripper or a carried object strikes the part row, the nest, the chip wall, the chip pan, a tool rest or the tray and knocks, tips, slides or displaces it; a part falls out of the nest or off the tray; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 6, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. The chip wall and the chip pan rim are the two things worth steering widest of.

Violation: Manipulation outside the camera frame

- **Visible cue:** a part is seated or trayed, an edge is filed or felt, or the nest is brushed, partly or wholly outside the environment camera frame, or the arm stands between the camera and the nest while a stroke, a pad pass or the felt is what has to be seen.
- **SOP rule broken:** Steps 1 to 6, every manipulation happens inside the camera frame, and the nest, the far edge and the felt stay visible.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions and keep the arm clear of the nest.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a part still on its stand or still in the nest, an edge unfiled or unfelt, a burr still standing on a trayed part, chips left on a part or in the nest, a part in the wrong pocket, a tool out of its home, the arm short of home, or the gripper not fully open.
- **SOP rule broken:** Step 8, confirm the row, the tray, the nest, the edges, the chip pan and the tools, then return the arm home with its gripper open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arm does before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective file: one that is bent, worn smooth in a patch, or whose teeth stay packed after a correct cleaning, so three correct strokes will not take the burr off.
- A file handle the gripper cannot close on squarely, or a file rest that will not hold the file in one fixed position teeth up.
- Defective check pad: felt that pulls threads on a clean edge, felt so tight that a known burr will not snag it, or a pad that comes loose off its stick.
- Defective brush: one whose bristles stay bent flat, or that sheds bristles into the nest.
- Defective part: one that is bent, whose top face is not flat, whose sides the gripper cannot close on squarely, or that will not sit flat in the nest or in a pocket.
- Defective nest: one that slides, turns or lifts under a correct stroke, one that leaves the part room to turn or slide, or a chip wall that falls over or stands so tall the file cannot ride over it.
- Defective tray: one that slides or lifts under a correct placement, or a pocket so tight that a part will not go down onto the floor of it or will not lift straight out.
- Defective part row: one that slides, turns or lifts under a correct pick, or a stand that grips a part so a straight lift will not free it.
- A chip pan that will not stay put, or whose rim stands high enough that a part carried correctly past it still touches it.
- Parts loaded already deburred, so there is no burr for the file to take off and no snag for the pad to find. Reset error, not a violation.

- A tray printed wrong: a missing pocket number, pocket numbers not running 1 to 4 left to right, or numbers turned away from the camera.
- Glare, shadow or a light flickering over the nest that stops a reviewer seeing the far edge or the felt. Fix the lighting before the next episode.
- Row holding fewer or more than four parts at episode start. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take one part from its stand
2. Seat one part in the nest
3. Take the file from its rest
4. File one edge with three strokes away
5. Return the file to its rest
6. Take the check pad from its rest
7. Drag one pad pass along one edge
8. Return the check pad to its rest
9. Turn one part a quarter turn in the nest
10. Take the brush from its holder
11. Sweep the chips into the chip pan
12. Return the brush to its holder
13. Lift one part out of the nest
14. Put one part in its tray pocket
15. Return the arm home and end the episode